A Deterministic Almost-Linear-Time Algorithm for the Maximum-Weight (k, ℓ)-Sparse Subgraph Problem

The full classical range and a uniform additive input bound

October 11, 2026

Abstract

For each fixed $k \geq 1$ and $0 \leq \ell < 2k$, we give a single deterministic word-RAM algorithm with worst-case running time $O(m+n^{1+o(1)})$ for the maximum-weight (k, ℓ)-sparse subgraph problem, assuming that the edges are already ordered by nonincreasing weight. The algorithm does not depend on an exponent-error parameter, and the coefficient of the m term does not depend on the recursion depth.

In the upper range $k < \ell < 2k$, we reduce the number of mandatory tight bodies geometrically using a fixed constant number of local greedy layers. Samples are generated implicitly from shared ordered incidence lists. Edges spanned by earlier selected edges of their body pair are deleted permanently. A separate saturation argument bounds priority-queue head repairs by $O(Dp + qD + f)$ at a stage with q bodies, p physical vertices, and f known auxiliary constraints. Fixed first bindings and activation times preserve chronological correctness across stages without repeated ancestor queries. The union of all samples is an $O(n)$-edge certificate.

This English version organizes the final construction and all required preliminary lemmas into a standalone presentation. The generic natural representation is used only to prove safety; the algorithm performs deterministic combinatorial operations.

Contents

1 Problem, model, and main theorem **1**
1.1 Algorithmic primitives 2

2 Tight sets and consecutive sparsity matroids **3**

3 Sparse optimization and the complete prefix history **4**
3.1 Rooted-cut reductions 4
3.2 Listing the circuits 5
3.3 All tight prefix blocks 5
3.4 A constant-time chronological closure index 7
3.5 The sparse optimizer 8

4 The lower range: an ordered union certificate **8**

5 Bodies, physical pins, and logical block size **9**
5.1 Frames and the natural representation 9
5.2 Logical size and the number of large bodies 11

6 Historical hinges, straddlers, and endpoint coverage **11**
6.1 The total historical hinge count 11
6.2 Old bodies spanning different large blocks 12
6.3 Logical ports and the potential 12

7 The adaptive local sample **14**
7.1 Cyclic layers of constant rank 14
7.2 A body-pair filter using only selected edges 14

8 Implicit execution on shared incidence lists **15**
8.1 Finding the next locally eligible edge 15
8.2 Saturations affecting many heads 16

9 The fixed first-binding interface **17**

10 One fixed algorithm and its total cost **19**
10.1 Fixing the constants once 19
10.2 The initial stage 19
10.3 The constant-body terminal stage 20
10.4 Correctness and the uniform additive bound 21

11 Exact checks and their scope **22**

1 Problem, model, and main theorem

Let $G = (V, E)$ be a loopless undirected multigraph with n vertices and m edges. Vertex identifiers belong to $\{1, \dots, n\}$. The input supplies a strict total edge order

$$e_1 \succ e_2 \succ \cdots \succ e_m$$

consistent with nonincreasing weight; input order breaks ties. Nonpositive-weight edges may be omitted without changing the maximum attainable weight. We therefore work with positive weights. All subsequent greedy-output and certificate equalities refer to this retained positive-weight input. An empty input is returned immediately; we may therefore assume $n \geq 2$.

Definition 1.1. An edge set I is (k, j)-*sparse* if

$$|I[X]| \leq \max\{k|X| - j, 0\} \qquad (X \subseteq V).$$

A set X with $|X| \geq 2$ is *tight* if $|I[X]| = k|X| - j$. A *block* is an inclusion-wise maximal nontrivial tight vertex set.

For $0 \leq j < 2k$, these independent edge sets form a matroid, denoted by M_j. Write $B_j(A)$ for the ordered matroid-greedy output on an edge set A, and $A_{\succ e}$ for the edges of A preceding e. The greedy output is a maximum-weight independent set and is determined by the supplied order.

Theorem 1.2. *For fixed $k \geq 1$ and $0 \leq \ell < 2k$, the maximum-weight (k, ℓ)-sparse subgraph problem on ordered edges has a single deterministic word-RAM algorithm with worst-case running time*

$$\boxed{O(m + n^{1+o(1)})}.$$

The program and the coefficient of the m term depend only on k, ℓ. No choice of algorithm indexed by an arbitrarily small positive exponent is required.

The word length is $\Omega(\log(n+m+2))$, with a sufficiently large constant to address the polynomial-size tables used below. Vertex and edge identifiers, root-prefix timestamps, and the polynomially bounded integers in the combinatorial data structures fit in a machine word. The constants and the $o(1)$ term may depend on fixed k, ℓ. We do not claim the same bound when k is part of the input, or a corresponding almost-linear space bound. In particular, constant-time initializable direct-address tables are allowed [1].

The flow subroutine in [3] uses fixed-point numbers with $\log^{O(1)} z$ bits on an instance encoded with z bits (its Section 2, "Computational Model"). On our sparse networks $z = N \log^{O(1)} N$. Simulating those arithmetic operations on ordinary machine words has a polylogarithmic overhead, which is absorbed in $N^{1+o(1)}$. This simulation is confined to the sparse subroutines; the operations charged directly to the original m records use only a constant number of machine-word operations.

The proof proceeds in three parts. First, we construct a deterministic almost-linear optimizer for sparse inputs and a compact history of all tight prefix blocks. Second, we establish a local sampling and body-compression lemma. Third, we execute sampling implicitly and compose the stages with fixed endpoint bindings. The complete algorithm appears in Section 10.

1.1 Algorithmic primitives

The following published algorithms are used as subroutines. All their parameters other than the instance size are fixed here.

1. Exact minimum-cost flow and circulation with polynomially bounded integer capacities and costs can be computed deterministically in almost-linear time [3].

2. An edge set known to decompose into k forests can be decomposed deterministically in almost-linear time [4].

3. A rooted unit-capacity digraph known to have rooted connectivity at least a admits a actual edge-disjoint spanning out-arborescences, computable in $\widetilde{O}(N+A)$ time for fixed a [6]. The bound reported in [7] is $O(Aa \log N + Na^4 \log^2 N)$ for exact rooted connectivity a. If only a lower bound is given, attach a new root by a parallel arcs to the old root, apply that algorithm, and remove these arcs.

4. For a fixed positive minimum-cut value, the required inclusion-wise minimal directed minimum-cut sets can be extracted in $\widetilde{O}(N+A)$ time from the minset-poset representation [8, 9]. For value zero, SCCs suffice.

Here N and A denote network vertices and arcs. Each explicit sparse instance is relabelled densely on $\{1, \dots, N\}$; its separate metadata retain the original vertex identities and timestamps. The reductions below use these algorithms only on sparse networks. The orientation and rooted-connectivity perspective also appears in the recognition algorithms of [5].

2 Tight sets and consecutive sparsity matroids

Lemma 2.1 (Uncrossing and closure)**.** *Let I be independent in M_j. If tight sets A, B have at least two common vertices, then $A \cap B$ and $A \cup B$ are tight, and there is no edge of I between $A \setminus B$ and $B \setminus A$. Distinct blocks therefore intersect in at most one vertex and have disjoint internal edge sets. For a new edge $e = uv \notin I$,*

$$e \in \mathrm{cl}_{M_j}(I) \quad \Longleftrightarrow \quad u, v \text{ belong to a common tight set.}$$

If $j < k$, distinct blocks are vertex-disjoint. Every nontrivial tight set has no isolated internal vertex.

Proof. Supermodularity of induced edge counts and sparsity give

$$k|A| + k|B| - 2j \le |I[A \cup B]| + |I[A \cap B]| \le k|A \cup B| + k|A \cap B| - 2j.$$

The two ends are equal, so all inequalities are equalities. For closure, a violated sparsity inequality in $I + uv$ must contain both endpoints and must have been tight in I. Conversely, adding an internal edge to a tight set violates its bound. With a one-vertex intersection, two tight sets would violate the union bound when $j < k$. For the last assertion, a tight two-vertex set has $2k - j > 0$ edges. In a larger tight set, removing an isolated vertex would leave the same edge count but lower the sparsity bound by k, a contradiction. □

Lemma 2.2 (Spanned tight graphs)**.** *Let I be independent in M_j, and let T be an independent tight graph on X, where $|X| \ge 2$. If every edge of T belongs to $\mathrm{cl}_{M_j}(I)$, then X is contained in a block of I.*

Proof. Let $\mathcal{Z}$ be the blocks of I meeting X in at least two vertices, and put $Y = X \cup \bigcup_{Z \in \mathcal{Z}} Z$. Every edge of T outside these blocks belongs to I, by Lemma 2.1. If $\mathcal{Z}$ is empty, this gives $T \subseteq I$, so X itself is tight in I. Otherwise the internal edge sets of the blocks are disjoint, and

$$\begin{aligned} k|X| - j = |T| &\le |I[Y]| - \sum_{Z \in \mathcal{Z}} |I[Z]| + \sum_{Z \in \mathcal{Z}} (k|X \cap Z| - j) \\ &\le k|Y| - j - \sum_{Z \in \mathcal{Z}} (k|Z| - j) + \sum_{Z \in \mathcal{Z}} (k|X \cap Z| - j) \\ &\le k|X| - j. \end{aligned}$$

The last inequality uses $|Y| \le |X| + \sum_Z (|Z| - |X \cap Z|)$. Equality throughout forces $|I[Y]| = k|Y| - j$. Thus Y is tight and a block containing Y contains X. □

Lemma 2.3 (Disjoint circuits)**.** *Let $1 \le j < 2k$. If I is independent in M_{j-1}, its M_j-circuits are pairwise edge-disjoint. Their supports are precisely the inclusion-wise minimal nontrivial $(k, j-1)$-tight sets W, and each such circuit is*

$$I[W], \qquad |I[W]| = k|W| - (j-1).$$

Consequently, $B_j(I)$ is obtained by deleting the last edge of each circuit in the supplied order.

Proof. Every edge of a circuit must be internal to a violated set; otherwise its internal edges give a proper dependent subset. Independence in M_{j-1} and dependence in M_j determine the displayed edge count. If two circuits shared an edge, their supports would meet in two vertices. Uncrossing would make their intersection weakly tight and hence dependent in M_j, contradicting minimality unless the circuits were equal. Every dependence contains a circuit, and disjoint circuits interfere with none of the other greedy deletions. □

Lemma 2.4 (Greedy compatibility)**.** *For every ordered E and $1 \le j < 2k$,*

$$B_j(B_{j-1}(E)) \subseteq B_j(E).$$

Proof. Suppose the stronger greedy algorithm accepts e when restricted to the weaker greedy basis. Let D be the weaker accepted prefix before e; thus $D + e$ is weakly independent. Let $\mathcal{Z}$ be the maximal weakly tight blocks of D. Every earlier edge rejected by the weaker algorithm is internal to a member of $\mathcal{Z}$. The stronger restricted greedy algorithm accepts every edge of D outside these blocks: otherwise Lemma 2.3 would give a weakly tight circuit support containing such an edge. The edge e is also outside the blocks.

Let A be the stronger full greedy prefix before e. Construct K from the edges of $D+e$ outside the blocks and the sets $A[Z]$, $Z \in \mathcal{Z}$. Then $A+e \subseteq K$. For $|X| \geq 2$, put

$$\mathcal{Z}_X = \{Z \in \mathcal{Z} : |Z \cap X| \geq 2\}, \qquad Y = X \cup \bigcup_{Z \in \mathcal{Z}_X} Z.$$

If $\mathcal{Z}_X$ is nonempty, weak sparsity of $D+e$ bounds the outside-block contribution by

$$k|Y| - (j-1) - \sum_{Z \in \mathcal{Z}_X} (k|Z| - (j-1)).$$

The inside contribution is at most $\sum_{Z \in \mathcal{Z}_X} (k|X \cap Z| - j)$. Using $|Y| \leq |X| + \sum_{Z \in \mathcal{Z}_X} (|Z| - |X \cap Z|)$, their sum is at most $k|X| - (j-1) - |\mathcal{Z}_X| \leq k|X| - j$. If $\mathcal{Z}_X$ is empty, $K[X]$ is a subset of the stronger restricted accepted prefix together with e, and is independent. Thus K is independent in M_j, so $A+e$ is independent as well. □

3 Sparse optimization and the complete prefix history

This section supplies the static algorithms used later on explicit graphs of $O(N)$ or $O(N \log(N+2))$ edges. These graphs contain only registered physical vertices and constant-size body frames; the original dense input is never passed to them.

3.1 Rooted-cut reductions

Let I be (k,t)-sparse. If $t \geq k$, decompose I into k forests $T_1, \ldots, T_k$. A tight set C satisfies

$$\sum_{i=1}^{k} (|C| - 1 - |T_i[C]|) = t - k.$$

Every summand is nonnegative, so at least one of the first $t-k+1$ forests is connected on C. If $0 < t < k$, decompose I into t forests and $k-t$ pseudoforests. Tightness implies connectivity of C in every forest part, so the first forest suffices. This mixed decomposition can also be constructed almost linearly: orient I with indegree at most k, add $k - d^-(v)$ root arcs to each vertex, and extract t edge-disjoint spanning out-arborescences. Removing their root arcs gives the forest parts; the remaining orientation has indegree at most $k-t$ and can be split into that many pseudoforests.

Centroid-decompose the selected forest components. For every connected tight C, some call has a vertex set U and centroid u with $u \in C \subseteq U$: take the first centroid in C. Over all calls, vertex-set sizes sum to $O(N \log(N+2))$. A fixed bounded-indegree orientation permits induced edges to be listed with the same bound by scanning at most k incoming edges per vertex and using membership marks.

For $t \geq k$, root every local forest at u, remove u, and add $k - d^-(v)$ arcs $r \to v$, retaining the indegrees measured before removing u. The resulting digraph R satisfies

$$\rho_R(X) = k|X| - |I[X \cup \{u\}]| \geq t - k =: \delta. \tag{1}$$

Nonempty δ-cut sets correspond exactly to tight sets containing u. For $0 < t < k$, orient locally with capacities $b(u) = k-t$ and $b(v) = k$ otherwise, and add root arcs to fill these capacities. Then

$$\rho_R(X) = k|X| - t\mathbf{1}_{u \in X} - |I[X]| \geq 0. \tag{2}$$

Every nonempty zero-cut contains u and is a tight set. Indeed, a root-avoiding set not containing u has cut value at least $t > 0$, while the singleton $\{u\}$ has cut value $k - t > 0$. The required orientation follows from capacitated Hall inequalities and is found by a sparse flow reduction. For $t = 0$, use one global bounded-indegree orientation and root completion; its cut formula is $\rho_R(X) = k|X| - |I[X]|$.

3.2 Listing the circuits

To compute Circuits(j, I), apply the reductions with $t = j - 1$. In each call, enumerate the inclusion-wise minimal nonempty root-avoiding cut sets of the target value (zero or δ), and translate them back to vertex sets. If there are no such sets, the call contributes no candidate. For positive δ, the rooted reduction to the global minset-poset problem adds δ arcs from every nonroot vertex to r, a new vertex z, and δ arcs in each direction between r and z. The global minimum-cut value is δ, and cuts on the original nonroot vertices are unchanged. Every cut containing r receives either the δ arcs from z or the δ arcs of a missing original vertex; a cut containing z but not r receives the δ arcs $r \to z$. Thus no smaller cut is introduced. Keep only minimal cuts avoiding both r and z. For value zero, use source SCCs having no entering arc from outside the SCC; the root SCC is excluded.

Minimal root-avoiding minimum-cut sets are disjoint: a nonempty intersection would, by cut submodularity, itself be a minimum-cut set. Candidate sizes therefore sum to $O(|U|)$ per call, even when the centroid is added to every candidate. Every true circuit support appears in its covering centroid call. Process candidates by increasing vertex-set size, accepting a candidate precisely when none of its internal edges has already been marked; mark those edges on acceptance. A nonminimal candidate contains a smaller true circuit. A candidate sharing an edge with an accepted circuit contains that circuit, by uncrossing and minimality. This procedure therefore lists exactly the circuits of Lemma 2.3, in $N^{1+o(1)}$ time.

3.3 All tight prefix blocks

Fix an independent ordered set $F = \{f_1, \dots, f_a\}$, and let $F_t = \{f_1, \dots, f_t\}$. If C is tight in a prefix, it is tight in the final set and $F[C] = F_t[C]$: a later internal edge would violate final independence. Its birth time is

$$b(C) = \max\{i : f_i \in F[C]\}.$$

Hence the forest and centroid decompositions of final F cover every prefix tight set.

Build each local rooted network from final F and delete original edges in reverse order. An arc deletion $a \to b$ is implemented as

$$a \to b \;\longmapsto\; r \to b. \tag{3}$$

This preserves the indegree total and can only increase root-cut values. In the upper reduction, deleting an edge from the removed centroid simply adds a root arc.

For zero cuts, the maximal zero-cut is the set unreachable from r. For reachability, keeping the old arc and adding the new root arc is equivalent to replacement. Each vertex becomes reachable at most once, giving linear total scanning work in a local call.

For $\delta > 0$, extract δ spanning out-arborescences $P_1, \dots, P_\delta$. After root arcs are removed, let $Q_1, \dots, Q_\delta$ be their forests, and let Z contain the other arcs. A nonempty X is a δ-cut exactly when each $Q_i[X]$ is connected and no Z-arc enters X: each arborescence contributes one entering arc per induced forest component, and all additional contributions are nonnegative.

Maintain a partition of live vertices, refining it until every part is connected in every Q_i. If an extra arc enters a part from outside, delete its head from the live set and propagate further refinements. Every possible δ-cut remains live and contained in one part: connectivity forces containment through refinements, and an entering extra arc excludes its head from every candidate cut contained in that part. At stability, the parts themselves are δ-cuts, hence are exactly the maximal ones.

If a deleted arc belongs to P_i, its subtree is reattached to the root. The arborescence remains valid, while Q_i loses one edge. If the deleted arc is extra, its replacement root arc excludes its head. Thus only splitting and deletion are needed throughout the reverse history.

Dynamic forests and smaller-side relabeling give a $\widetilde{O}(N + A)$ total bound for each local history. Here is an implementation of the refinement and propagation work, including arcs that become external when a part splits.

Maintain each Q_i as a decremental forest, cutting also every forest edge whose endpoints acquire different part labels and every edge incident to a deleted vertex. These additional cuts do not remove an edge internal to any surviving candidate cut: such a candidate is contained in one current part and uses only live vertices. Consequently every maintained forest component is contained in one part. Keep an explicit component identifier for each live vertex, retaining the identifier of the larger component on a cut and relabelling the smaller component. Deterministic balanced Euler-tour trees give component sizes, cuts, and enumeration of the smaller component in $O(\log(N + 2) + \text{enumerated size})$ time. Each vertex changes its component identifier only $O(\log(N + 2))$ times in each of the fixed number of forests.

For every part and every forest, store a balanced dictionary of the component identifiers occurring in that part, with their vertex counts. If two components occur, enumerate the smaller of two such components and move it to a new part. It has at most half the old part's vertices. Update the dictionaries and scan the original forest and extra-arc incidences of moved vertices. Cut forest edges made external by the split, and queue the head of each extra arc newly entering a part. Scan all extra arcs once initially. When a queued head is deleted, cut its forest incidences, remove it from the dictionaries, and scan its extra incidences; in particular an outgoing extra arc may now exclude its live head. Duplicate queue entries are checked against the current labels and live flags before they are used. Perform all cross-part forest cuts before choosing the next refinement.

Part labels, like forest-component labels, move only to smaller classes, so a live vertex changes them $O(\log(N + 2))$ times. Each original incidence is therefore scanned $O(\log(N + 2))$ times, each forest edge is cut at most once, and each vertex is deleted at most once. Dictionary and forest operations add at most polylogarithmic factors. This proves the stated $\widetilde{O}(N + A)$ total bound without charging a scan of a surviving large part after every update.

Record stable parts in a splitting/deletion tree, suppressing unchanged parts and storing vertex membership by the tree rather than copying every part. Splits have linear total tree size; unary shrinkages are charged to the vertices deleted at those shrinkages. Thus each local history has linear size.

To compile the local trees globally, assign each original edge to its smallest local tight ancestor, if one exists. This ancestor is found from the endpoint leaves by LCA preprocessing of the membership tree; in the upper reduction an edge incident to the centroid uses the other endpoint's leaf. The centroid is common to all the local tight sets and is not duplicated as a membership leaf. Ancestors that do not represent stable tight sets are skipped. Internal edge sets are laminar within a local tree, so births are computed by subtree maxima of assigned edge indices. These assignments and computations take $\widetilde{O}(N + A)$ time. Form a timestamped auxiliary graph on original edge-elements and local tight nodes: connect each edge to its assigned node at that node's birth, and each node to its first tight parent at the parent's birth.

At time t, retain only active components containing at least one activated tight node. Their projections onto original edge-elements are exactly the internal edge sets of the maximal blocks of F_t. Indeed, a shared internal edge permits tight-set uncrossing and joins only one global block. Conversely, every global block is a maximal local candidate in a centroid call covering it, so all its internal edges are connected there. An isolated edge leaf, or an edge group with no activated tight node, is not a block. This distinction is necessary, for example, when a single accepted edge is not yet a tight two-vertex graph.

Process equal timestamps in batches and record the resulting union history. Suppress auxiliary nodes and unary nodes that do not change an edge group. Preserve the first activation of a one-edge tight block separately from its edge leaf. There are $O(a)$ genuine block nodes and merges. A nontrivial tight set has no isolated internal vertex, so its vertex set is determined by

its internal edge endpoints. This yields the compact history in $N^{1+o(1)}$ time.

3.4 A constant-time chronological closure index

Orient final F with indegree at most k. For a historical tight block C, define its ports by

$$\Pi(C) = \{v \in C : d^-_{F[C]}(v) < k\}.$$

Since the internal indegree deficit sums to j, $|\Pi(C)| \le j$. A nonport vertex is *saturated*. It can be saturated in only one current maximal block, because those blocks have disjoint internal edge sets.

Compute port lists by merging internal indegree maps up the edge-group hierarchy, always moving the map of the smaller edge group, measured by its number of edge leaves. Each map has at most twice that many vertex entries, and the edge-group size at least doubles whenever it is moved. This gives $O(N \log(N+2))$ total entry visits even when maps share keys. Store the deficient entries separately, so an entire large map is never rescanned at every ancestor. Update this list only at touched keys; at a genuine tight node it has at most j entries. Deterministic dictionaries give total cost $\widetilde{O}(N)$. For a vertex of final indegree k, let its anchor be the lowest genuine block node containing all its incoming edge leaves. The anchor is found by a constant number of LCA queries in preprocessing, followed by the nearest genuine-block ancestor lookup. If the incoming leaves have no common genuine block ancestor, the vertex has no anchor; an artificial root joining different history trees is never a block. At time t, its highest activated ancestor is the saturated owner $o_t(v)$, if one exists. Explicit smaller-class relabeling on this known union tree costs $O(N \log(N+2))$ in total and permits $O(1)$ owner queries; the specialized union-find of [10] is another implementation.

For each distinct pair of ports, precompute

$$\beta(u, v) = \min\{b(C) : u, v \in \Pi(C)\},$$

with value $+\infty$ when the pair is absent. There are $O(N)$ such pairs. Two stable counting sorts by endpoint identifier, followed by merging with the query-pair list, give every offline β answer in $O(N + Q)$ time. For a new edge $uv \notin F_t$, closure holds exactly when

1. $o_t(u) = o_t(v) \neq \perp$;
2. one endpoint has a saturated owner and the other is a port of that owner; or
3. $\beta(u, v) \le t$.

These are precisely the cases in which both endpoints lie in a common current block. Each query uses constant work. For $j = 0$, closure is membership of both endpoints in the global unreachable set, whose entry times were recorded above.

Consequently, for fixed independent F, the operation

$$\textsc{Filter}(j, F, E) = \{e \in E : e \notin \mathrm{cl}_{M_j}(F_{>e})\}$$

costs $O(N^{1+o(1)} + |E|)$ deterministically. The block history and port lists themselves are computed without scanning a separate dense edge set.

3.5 The sparse optimizer

Proposition 3.1. *For fixed $k \ge 1$, $0 \le j < 2k$, and $|E| = O(N \log(N+2))$, the ordered greedy output $B_j(E)$ can be computed deterministically in $N^{1+o(1)}$ time. Mandatory independent edges may be placed first in the order.*

Proof. For $j = 0$, independence is equivalent to assigning each edge to one endpoint, with at most k assignments per vertex. Build a circulation network with an edge-node x_i for each ordered edge $e_i = uv$, arcs $s \to x_i$ of capacity one and cost $-(a - i + 1)$, zero-cost unit arcs $x_i \to u, x_i \to v$, arcs $v \to t$ of capacity k, and a zero-cost return arc $t \to s$ of capacity kN, where $a = |E|$. An integral minimum-cost circulation selects the maximum weight independent set for these strictly ordered rank costs. Matroid greedy depends only on the order, so its output is exactly $B_0(E)$ for the original weights as well. The distinct positive surrogate weights make the optimal edge set unique; alternative optimal routings of its endpoint assignments do not affect the selected set. All numbers are polynomially bounded, and [3] gives the required almost-linear running time.

For $j > 0$, repeatedly perform

$$\begin{aligned} &B \leftarrow \textsc{Sparse}(j-1, E),\\ &F \leftarrow B \setminus \{\text{last edge of each } C \in \textsc{Circuits}(j, B)\},\\ &E \leftarrow \textsc{Filter}(j, F, E), \end{aligned}$$

until $E = F$. Initially take the previous round's set to be empty. The filter retains F, because F is independent. The filter never deletes an edge of the full greedy optimum O: a deleted edge is spanned by earlier input edges. By Lemmas 2.3 and 2.4, every produced F is contained in O.

Let F_0 be the previous round's set. Deleting input edges cannot turn a previously accepted weaker greedy edge into a rejected edge, so $F_0 \subseteq B$. Since $F_0 \subseteq O$, the stronger restricted greedy algorithm also accepts its edges, and $F_0 \subseteq F$. Every stronger circuit in B contains at least two edges outside F_0. Its last edge cannot belong to F_0. If that last edge were its only edge outside F_0, the previous filter would already have deleted it. The circuits are edge-disjoint, so at most $|B \setminus F_0|/2$ edges are deleted. With $r = |O|$,

$$|F| - |F_0| \geq \frac{|B| - |F_0|}{2} \geq \frac{r - |F_0|}{2}.$$

The missing optimum size halves every nonfinal round. There are $O(\log(N+2))$ rounds. Once $F = O$, the next filter leaves exactly F.

The recurrence is $T_0(N) = N^{1+o(1)}$ and $T_j(N) \leq O(\log(N+2))(T_{j-1}(N) + N^{1+o(1)})$. Its depth is fixed, so $T_j(N) = N^{1+o(1)}$. Independent mandatory edges placed first are all accepted. □

4 The lower range: an ordered union certificate

For $0 \leq \ell \leq k$, the sparsity matroid is the union of ℓ graphic and $k - \ell$ bicircular matroids; the latter have pseudoforests as their independent sets [5]. We use the following general certificate.

Lemma 4.1. *Let $U = N_1 \vee \cdots \vee N_k$ be a union of ordered matroids on E, and let R be an integer with $R \geq \max\{1, \sum_i r_i(E)\}$. Set $L = 1 + \lceil \log_2(R+1) \rceil$. Create L cycles of layers, each cycle containing $N_1, \ldots, N_k$ in that order. Insert each edge into the first layer that accepts it, or discard it if all reject. The union H of the layers contains $B_U(E)$, and $|H| \leq RL$.*

Proof. Every earlier layer spans every element stored in a later layer in the earlier layer's matroid, at every input prefix. Suppose U-greedy accepts e but every layer rejects it. Put $A = E_{>e}$ and choose a minimizer $X \subseteq A$ of the union rank formula

$$r_U(A) = |A \setminus X| + \sum_i r_i(X).$$

Write $Q_i = N_i/X$. Comparing X with $X \cup Y$ shows that every $Y \subseteq A \setminus X$ satisfies $|Y| \leq \sum_i r_{Q_i}(Y)$. Thus $A \setminus X$ is independent in the union of the quotients and has size at most R.

Moreover, e is not a loop in at least one quotient; otherwise choosing $X + e$ in the rank formula for $A + e$ would show that e does not increase union rank.

All layer sets in the rest of this argument are evaluated just before e. Let K_c be the stored elements outside X in cycles strictly after cycle c. Each layer $B_{c,i} \setminus X$ spans K_c in Q_i. Hence

$$|K_c| \le \sum_i r_{Q_i}(K_c) \le \sum_i |B_{c,i} \setminus X| = |K_{c-1}| - |K_c|.$$

It follows that $|K_c| \le |K_{c-1}|/2$. Since $|K_0| \le R$, we have $|K_{L-1}| < 1$. Every layer in the final cycle is therefore contained in X, yet spans e. This makes e a loop in every Q_i, a contradiction. Thus no greedy optimum edge is discarded. Removing greedy-rejected edges does not change the greedy output, and each cycle has size at most R. □

Proposition 4.2. *The lower range has a deterministic $O(m + n^{1+o(1)})$ algorithm.*

Proof. Use Lemma 4.1 with $R = kn$. Within each layer, store explicit component identifiers and component lists, relabelling the smaller list on a merge. A graphic layer rejects uv exactly when its endpoints are in the same component. A bicircular layer rejects uv exactly when both endpoints belong to unicyclic components, which may be the same component. Tree-component vertices are marked when their component becomes unicyclic; every vertex receives this mark at most once in a layer. Closure queries therefore take worst-case constant time, and maintenance costs $O(n \log(n + 2))$ per layer.

Within each of the two layer types, earlier layers span later layers, so rejecting layers form an initial segment. First test only the final layer of each type that is present. If every present type rejects, discard the edge in constant time. Otherwise binary-search the accepting types to find the first accepting layer, and choose the earliest among them. Binary searches occur only for retained edges, of which there are $O(n \log(n + 2))$. The total construction cost is $O(m+n \log^2(n+2))$. Apply Proposition 3.1 to the resulting $O(n \log(n+2))$-edge certificate. □

5 Bodies, physical pins, and logical block size

For the remainder of the construction, assume $k < \ell < 2k$ and set $c = 2k - \ell \ge 1$. A *body* is a mandatory tight graph. Its *pins* are physical vertices, whose identities are retained even when the internal body graph is replaced by a small frame. Different bodies at the start of a stage are vertex-disjoint. Their mandatory bases have union P_0.

5.1 Frames and the natural representation

The natural representation of the sparsity matroid [2] can be described using a generic matrix U with k rows U_v per physical vertex and ℓ columns. An edge row is supported on its endpoints and satisfies $a_{e,u}U_u + a_{e,v}U_v = 0$. The nullspace of a tight body's internal basis is exactly

$$\xi_v = U_v \alpha_A \quad (v \in A), \qquad \alpha_A \in \mathbb{R}^\ell. \tag{4}$$

The stacked matrix U on a nontrivial body has column rank ℓ, and its independent tight basis has $k|A| - \ell$ rows. Its nullspace therefore has dimension ℓ and equals the displayed ℓ-dimensional space, which is already annihilated by every internal row. In particular the body parameter is unique. All generic statements can be made simultaneously on a sufficiently large universe, padded with isolated vertices if necessary. No such matrix is constructed by the algorithm.

A constant-size tight frame can be built explicitly. Let $a = \max\{2, \lceil \ell/c \rceil\}$. On a vertices, the complete multigraph with c copies per pair contains a tight graph with $ka - \ell$ edges. For proper subsets of size $x \ge 2$,

$$kx - \ell - c\binom{x}{2} = -\frac{c}{2}(x - 2)(x - \ell/c) \ge 0.$$

Retain any $ka - \ell$ edges. Enlarge the frame by adding vertices with k incident edges, at most c per old vertex pair; $ca \geq k$ guarantees sufficient capacity. This preserves sparsity and tightness. Choose the final constant frame size at least

$$h = \max\{k, 2, \lceil \ell/c \rceil\}. \tag{5}$$

Attach a fresh pin v to k distinct frame vertices. The enlarged graph is tight, so these attachments impose exactly $\xi_v = U_v \alpha_A$. Pins need only be registered when an actually retained edge uses them. Before registration, their private attachments are understood implicitly. Registering a fresh pin creates no new maximal block apart from extending the block containing its frame. To see this, let C be a maximal tight block containing the fresh pin before any additional incident edge is inserted. A nontrivial tight set has no isolated internal vertex, so C contains at least one of the pin's frame neighbors. It therefore meets the enlarged tight frame in at least two vertices. Uncrossing and maximality force C to contain the whole enlarged frame. Smaller tight subsets need not contain all the attachments: for $(k, \ell) = (2, 3)$ a single attachment is already tight on its two endpoints.

Lemma 5.1 (Exact disjoint-body count)**.** *For an edge set J joining distinct bodies, let $p_i(S)$ be the number of distinct pins in body A_i incident to $S \subseteq J$. Then J is independent after contracting P_0 exactly when*

$$|S| \leq \sum_i \min\{\ell, kp_i(S)\} - \ell \qquad (\varnothing \neq S \subseteq J). \tag{6}$$

Proof. For necessity, include the whole A_i when $p_i(S) \geq 2$, and include only the one incident pin when $p_i(S) = 1$. Subtracting the mandatory internal edges from the sparsity bound on this vertex set gives (6).

For sufficiency, take any vertex set Y and put $a_i = |Y \cap A_i|$. If $J[Y] \neq \varnothing$, the internal mandatory contribution is at most $ka_i - \ell$ for $a_i \geq 2$, and zero for $a_i = 1$. Apply (6) to $J[Y]$:

$$|P_0[Y]| + |J[Y]| \leq \sum_{a_i \geq 2} (ka_i - \ell) + \sum_i \min\{\ell, ka_i\} - \ell = k|Y| - \ell.$$

If $J[Y]$ is empty, mandatory sparsity is enough. □

Corollary 5.2 (Constant-rank local oracle)**.** *Between two disjoint bodies, J is independent exactly when*

$$|J| \leq \ell, \qquad d_J(v) \leq k \ (v \ a \ pin), \qquad |J_{uv}| \leq c \ (u, v \ a \ pin \ pair). \tag{7}$$

Independence and closure relative to a basis of size at most ℓ are decidable in constant time.

Proof. In (6), each body's contribution is k for one incident pin and ℓ for at least two. Subsets using at least two pins on both sides have bound ℓ; a star has bound k; and a single pair has bound c. These are exactly the displayed conditions and their subset consequences. □

For a fixed body A_i, replace all its neighboring bodies by one independent body, retaining every physical neighbor pin identity. This gives its *local matroid*, of rank at most ℓ, with oracle (7). This operation is used for sampling, not as a global independence oracle.

5.2 Logical size and the number of large bodies

Let F be an independent set of new edges after P_0. Its size is at most ℓq, where q is the number of old bodies. For $q \geq 1$, sparsity on the union of all body vertices and subtraction of their mandatory bases gives even $|F| \leq \ell(q - 1)$; for $q = 0$, F is empty. Throughout the block history all of P_0 is present. Prefixes during the preliminary insertion of individual mandatory edges are

not part of this history; they are contracted to the single initial timestamp zero. In a prefix $P_0 + F'$, a maximal tight block C either contains a whole A_i or meets it in at most one pin, by uncrossing. Let b_C count the whole old bodies in C and p_C the remaining pins. Define its *logical size* by $\sigma(C) = b_C + p_C$; this is also the number of old bodies it meets.

Lemma 5.3. *For every prefix block,*

$$|F'[C]| = \ell b_C + k p_C - \ell. \tag{8}$$

The sum of logical sizes of current blocks is $O(q)$. For $\sigma(C) \geq 2$, $|F'[C]| \geq (c/2)\sigma(C)$. For a threshold $s \geq 2$, the number Q of persistent identifiers born for blocks of logical size at least s satisfies

$$Q \leq \frac{2\ell q}{cs}. \tag{9}$$

Proof. Subtract the internal mandatory bases of the b_C complete bodies from tightness to obtain (8). Let r be the number of current blocks containing whole old bodies. They partition those q bodies. All current internal new-edge sets are disjoint, so

$$k \sum_{b_C \geq 1} p_C + \sum_{b_C = 0} (k p_C - \ell) \leq \ell q - \ell(q - r) = \ell r.$$

In the second sum, $p_C \geq 2$ and $k p_C - \ell \geq c p_C/2$. The total pin contribution is therefore $O(q)$. Also $\ell b_C + k p_C - \ell \geq k\sigma(C) - \ell \geq c\sigma(C)/2$ when $\sigma(C) \geq 2$.

Whenever a new large block contains a previous large block, inherit one such identifier and retire the others. Otherwise create a new identifier and reserve the new block's internal F-edges. Birth blocks of different identifiers have disjoint such edge sets: a shared edge would force containment in an existing large descendant. Every birth reserves at least $cs/2$ edges, proving (9). □

The compact prefix history from Section 3 computes these objects in $q^{1+o(1)}$ time on the actually registered source graph. Vertices not incident to F remain lazy. A lazy pin can enter a maximal block only together with its whole old body.

6 Historical hinges, straddlers, and endpoint coverage

6.1 The total historical hinge count

Lemma 6.1. *Consider a tracked family of tight sets in an independent monotone graph. Active sets intersect pairwise in at most one physical vertex. An identifier follows an increasing set until it is retired into a containing tracked set. Let R be the total number of identifiers ever introduced. The set T of vertices ever belonging to at least two active tracked sets has*

$$|T| \leq 3\ell R.$$

After identifying retired body parameters with their successors, all required point–body attachments and identifier identifications have an $O(R)$-size history.

Proof. Fix a bounded-indegree orientation of the final independent graph. Each tracked tight set has at most ℓ ports, because its internal indegree deficit is ℓ. A shared vertex is a port of at least one participating set: disjoint internal edge sets cannot both contain all its k incoming edges. Thus at most ℓR vertices are shared at any one time.

A shared vertex ceases to be shared only when its participating identifiers are identified. At an identification involving $a \geq 2$ identifiers, each disappearing shared vertex is a port of one of those sets, so at most ℓa disappear. There are at most R retirements, and $\sum a \leq 2R$, charging

each event to its $a-1$ retired identifiers. The number of first appearances as a shared vertex is at most the final shared count plus all disappearances. Hence $|T| \le 3\ell R$.

For the attachment history, track port occurrences $(v, \text{identifier})$ with $v \in T$. A retired identifier loses at most ℓ such occurrences. On a surviving increasing identifier, a port disappears only by becoming saturated. The first saturated owner of a vertex persists in a growing successor set. A different saturated owner would require identification: two simultaneously distinct tight sets cannot both saturate that vertex. The first saturation is charged once per vertex, and the other losses to retirements. The total number of port occurrences is therefore $O(\ell R+|T|) = O(R)$.

Record an attachment for each new port occurrence. For a saturated occurrence, an attachment to its first saturated owner suffices; later parameter identifications transfer it to successors. This records all necessary relationships. The prefix history, port lists, and owner updates extract them in almost-linear preprocessing time. The argument uses tightness, pairwise one-vertex intersection, and monotone containment, rather than a particular size threshold. □

6.2 Old bodies spanning different large blocks

An old body is a *straddler* if it is not wholly contained in a large current block but has at least two distinct pins in large blocks. Those pins lie in different large blocks.

Lemma 6.2. *With b current large blocks and a current straddlers,*

$$ca \le \ell(b-1) \qquad (b \ge 1).$$

If Q large identifiers are born in the complete history, the set X_{h} of old bodies ever becoming straddlers satisfies

$$|X_{\mathrm{h}}| \le (2\ell/c + 3\ell)Q. \tag{10}$$

Proof. Take the internal bases of the b large blocks and the a straddling old bodies. They are edge-disjoint subsets of the independent source graph. Counting their vertex sets separately overcounts their union by at least $2a$. Sparsity of the union gives $2ka \le \ell(b+a-1)$, which is the first inequality.

A straddler loses that status only when swallowed entirely by a large block; it can never become a straddler again. If the union of d old large children of the swallowing block already contains two pins of the swallowed body, the snapshot bound charges at most $\ell(d-1)/c$ such deaths to the children. The sum of $d-1$ over large identifications is at most Q.

Otherwise one of its former large-block pins z lies in a large block outside the swallowing union. After the swallowing, z is a shared vertex of two large blocks. Different swallowed old bodies give different z, because old bodies are disjoint. Lemma 6.1 therefore bounds these deaths by $3\ell Q$. Finally at most $\ell Q/c$ straddlers remain alive. These three bounds give (10). The second death case is essential: swallowing a straddler need not merge two preexisting large blocks. □

6.3 Logical ports and the potential

Assign each edge of final F to one of its physical endpoints so that a pin receives at most k assignments and all pins of one old body together receive at most ℓ. Such an assignment follows from Lemma 5.1 and capacitated Hall inequalities. It is found by an almost-linear sparse flow computation. Explicitly, use a unit-capacity source arc for each edge of F, arcs from that edge-node to its two pin-nodes, capacity-k arcs from each pin-node to its old-body node, and capacity-ℓ arcs from body nodes to the sink. For every $S \subseteq F$ the available endpoint capacity is $\sum_i \min\{\ell, kp_i(S)\}$, at least $|S|$ by (6). Capacitated Hall therefore routes all edge units. Integral flow gives the asserted assignment on an $O(q)$-size network of actually registered pins.

The logical objects of a block C are its complete old bodies, of capacity ℓ, and its individually intersected pins, of capacity k. By (8), their internal assignment deficit sums to ℓ. At most ℓ objects are therefore *logical ports*. This assignment is distinct from the full-graph orientation used in Lemma 6.1.

Fix $D \geq 4\ell$ and

$$s = \max\{2, \lfloor D/(4\ell) \rfloor\}.$$

Let $\mathcal{S}$ be the inclusion-wise maximal historical blocks of logical size less than s. Their internal new-edge sets are disjoint, and Lemma 5.3 gives

$$\sum_{C \in \mathcal{S}} \sigma(C) \leq q + 2\ell q/c.$$

Blocks of logical size one are old bodies and contribute no internal new edges. Define

$$p(A_i) = \deg_F(A_i) + \ell \sum_{\substack{C \in \mathcal{S} \\ A_i \text{ has a logical port object in } C}} (\sigma(C) - 1). \tag{11}$$

If A_i is only partially in C, its object is the single common pin. Since each block has at most ℓ logical ports,

$$\sum_i p(A_i) \leq P_* q, \qquad P_* = 2\ell + \ell^2(1 + 2\ell/c). \tag{12}$$

Let

$$X = \{A_i : p(A_i) \geq D/2\} \cup X_{\mathrm{h}}.$$

Equations (9)–(12) imply $|X| = O(q/D)$ and $Q = O(q/D)$.

Lemma 6.3 (Endpoint coverage)**.** *Suppose a chronological sample H has at most ℓ edges per old body pair, F is its greedy basis after P_0, and every omitted edge that remains globally open is spanned by each of D disjoint local layers at both end bodies. Then every such omitted edge has each endpoint in an old body from X or in a current large block.*

Proof. Let $u \in A_i$, with $A_i \notin X$, and suppose u is outside all current large blocks. If A_i were wholly in a large block, this would be impossible. Since A_i never straddles, at most one of its pins, say x, belongs to large blocks. Thus $u \neq x$.

Consider earlier sample edges at A_i. Edges in F are charged to $\deg_F(A_i)$. Every other earlier sample edge is spanned by its F-prefix, so both endpoints lie in a prefix block. A small such block lies in a member C of $\mathcal{S}$. The body-pair cap bounds its contribution at A_i by $\ell(\sigma(C) - 1)$. If its object is a logical port, this is charged to (11). Among the maximal historical small blocks in $\mathcal{S}$, there is at most one in which its object is saturated: full-body saturation uses ℓ assignments, and two saturated single pins would use $2k > \ell$ assignments on disjoint internal edge sets. That one small block contributes at most $\ell(s-1) \leq D/4$. Edges spanned in large blocks use pin x at A_i. Hence fewer than $p(A_i) + D/4 < 3D/4$ earlier sample edges can use a pin other than x.

Every one of the D disjoint local bases spanning the omitted uv must contain an edge not incident to x. Indeed, a basis using only x has size at most $k < \ell$, has degree zero at u, degree at most $c < k$ at v, and multiplicity zero on uv. The local oracle therefore does not span uv. This requires at least D non-x edges, a contradiction. The argument is symmetric for the other endpoint. □

7 The adaptive local sample

At a stage, let q be the number of disjoint mandatory old bodies. Besides remaining original edges, a sorted stream of $f = O(q)$ known auxiliary edges may arrive. These *known constraints* will be justified by earlier root-certificate edges in Section 9. Within the source graph they are ordinary chronological edges with distinct element identifiers.

7.1 Cyclic layers of constant rank

For each old body keep D local independent sets $B_1, \dots, B_D$. An incoming edge goes into the first local layer not spanning it, if there is one. At every prefix,

$$\mathrm{cl}(B_1) \supseteq \cdots \supseteq \mathrm{cl}(B_D). \tag{13}$$

An edge inserted into a later layer was already spanned by every earlier one, proving this invariant. Equal-rank nested flats coincide. Since local rank is at most ℓ, group consecutive equal-rank layers into at most $\ell + 1$ intervals, retaining a representative basis and each layer's actual assigned edges. Test interval representatives and update the first layer of the first accepting interval. Only its basis changes; split the interval and merge equal-rank neighbors. This costs $O_{k,\ell}(1)$ per processed edge and $O(D)$ initial storage per body.

7.2 A body-pair filter using only selected edges

For each old body pair (a, b), maintain H_{ab}, the selected sample edges on that pair. The chronological reference algorithm on $e = uv$, $u \in A_a$, $v \in A_b$, is the following.

1. If $a = b$, omit e as already spanned by P_0. If H_{ab} spans e relative to the two mandatory bodies, omit it safely.
2. Otherwise process e in the cyclic local layers of both endpoint bodies.
3. Select e into H and H_{ab} if either endpoint layer accepts it, or if it is a known constraint. Otherwise leave it outside H.

The pair test uses the exact oracle (7). It involves only earlier *selected* edges, rather than an independently scanned basis of the full input pair.

Lemma 7.1. *In every prefix, H_{ab} is independent and has size at most ℓ, and*

$$|H| \le \ell q D + f.$$

Let F be the chronological greedy basis of H after P_0. Every known constraint is spanned by P_0 and the F-prefix including its own source position. A known constraint omitted by the pair filter is already spanned by the strictly preceding F-prefix. Every omitted original edge that is not spanned by its $P_0 + F$-prefix is spanned by all D local layers at both endpoint bodies.

Proof. The pair filter inserts only rank-increasing elements into H_{ab}. Each selected original edge uses a local layer slot at an endpoint; there are $\ell q D$ slots. At most f additional known edges are selected. An omitted known edge, or an original edge rejected by the pair filter, is spanned by earlier H_{ab}. The relevant F-prefix is a basis of the H-prefix, so it spans these edges too. Any remaining omitted original edge was rejected in every local layer at both endpoints; all those layer edges belong to H. The bases are disjoint within one endpoint's layer system. □

This proves both hypotheses of Lemma 6.3. Known constraints are included in H or its prefix closure, and $|F| \le \ell q$ still holds. No full preliminary scan by body pair is required.

8 Implicit execution on shared incidence lists

Initially retain only the first c original edges of each physical vertex pair. The first c copies form a two-vertex tight set, so this preserves every ordered greedy prefix. A constant-time initializable direct-address pair table gives $O(m + n)$ preprocessing [1]. Build the original ordered incidence lists once, using doubly linked nodes, one at each endpoint of an edge. Unlinking the two nodes of an edge takes $O(1)$ time. All stages share these lists; they are never copied.

At a stage, each physical vertex has a view cursor. Its view starts at the current live list head and can advance past locally rejected edges without removing them from the shared list. Those edges remain available at the next stage. Original selected edges and original edges with a valid prefix-spanning certificate are removed permanently. Known auxiliary edges use separate small ordered lists. Stage membership and activation predicates are fixed; their safety is proved in Section 9.

8.1 Finding the next locally eligible edge

Lemma 8.1. *For a nonfull final local layer B_D and a fixed own pin u, its next locally eligible original incidence is found with constant local skipping work, apart from edges deleted permanently. A saturated own pin is disabled as a whole. Changes in B_D cause at most ℓ full point-head rebuilds per body.*

Proof. If $|B_D| = \ell$, disable local search for the entire body. Otherwise, an edge is locally rejected exactly because its own pin has degree k, its neighbor pin has degree k, or its physical pair has multiplicity c. There is at most one saturated own pin because $|B_D| < \ell < 2k$; disable its head in constant time. For another own pin u, at most ℓ neighbor identifiers are forbidden. Each has at most c original incidences with u in the entire root input. Thus at most $c\ell$ locally forbidden original records are skipped before the next eligible record. In fact this also bounds their total number at one own pin throughout a stage before that pin or its body is disabled: B_D only gains edges, its union of neighbor identities has size at most ℓ, and a view cursor never moves backwards.

The last layer grows at most ℓ times, and its flat is unchanged between growth events. On each growth, recompute all point heads in that body, continuing from the current view cursors rather than resetting them. The separate known-edge lists contain $O(f)$ records; their total cursor advancement is $O(f)$. □

Keep an inner heap of eligible point heads for each body and an outer heap of the body minima. Compare the outer minimum with the next known constraint. Process the earliest edge that is locally eligible at some endpoint and still open in its body pair, or the next known constraint. When an endpoint's last layer grows, rebuild that body's point heads. Otherwise advance only the affected endpoint's head.

Pair closure may make a queued head stale. Repair it lazily: remove a whole chain of currently pair-spanned incidences, perform all their constant-time unlinks, and insert only the next valid head into the heap. There is no individual heap update for each deleted edge. If the other endpoint's view cursor points at the removed node, advance it in constant time. A queued copy at that endpoint may remain stale until its own later repair. Keep at most one queued record per point and keep its key separate from the live view cursor. A stale queued node is detected by its live flag; repairing it starts from the already advanced cursor. Thus deletion does not require updates of other points' heap entries, or traversal of chains of dead nodes.

The order of tests during head generation is essential. At either endpoint, check the static safety predicates of the entire edge (including both activation thresholds and both body identifiers), then local closure, and use the current H_{ab} test only on a locally eligible record. An earlier locally skipped record is never deleted using a later sample edge as a witness. Local closure only grows, so that record cannot become eligible again in this stage. Every still-live eligible record deleted by a pair test occurs after the already processed sample edges. Pair deletion therefore always has a chronological certificate.

8.2 Saturations affecting many heads

Lemma 8.2. *While an old body A_a has a nonfull final local layer, at most $\ell D + 1$ different pair bases H_{ab} can reach rank ℓ. At most $\ell D + \ell + 1$ different neighbor pins y can reach degree k in their pair base with A_a.*

Proof. Take the final still-nonfull basis B of the last layer during this active interval: immediately before its first full-rank insertion, or at the end of the stage if it never fills. A full pair basis has rank ℓ in the local matroid as well: on that one pair the local and pair oracles coincide. It contains an edge outside $\mathrm{cl}(B)$. Since layer flats only grow, that edge was accepted in a local layer of A_a on arrival. Different body pairs give distinct witnesses. There are at most ℓD local acceptances; the final edge that disables the body adds at most one event.

A degree-k pair star at a neighbor pin y either contains an edge outside $\mathrm{cl}(B)$, giving another local acceptance, or is wholly spanned by B. Stars for different y have disjoint edges. In the latter case y occurs in B. Indeed, all k star edges cannot use the single saturated own pin, because each physical pair has at most $c < k$ selected copies. A spanned star edge from another own pin therefore requires saturation of y in B or saturation of its physical pair in B. Both make y a neighbor endpoint of a basis edge. There are at most ℓ such y. Again the disabling edge adds at most one event. □

Lemma 8.3 (Implicit sampling cost)**.** *Let p be the number of physical vertices handled at a stage, and let d be the number of original edges permanently unlinked at that stage. The implicit procedure produces exactly the reference sample and local layers, in time*

$$O(d + (Dp + qD + f)\log(n + 2) + f). \tag{14}$$

For the fixed D used below, the coefficient of d does not depend on the number of stages.

Proof. Pair closure changes only on four kinds of saturation: total rank ℓ, own-pin degree k, neighbor-pin degree k, and physical-pair multiplicity c. The first and third kinds may affect all p_a point heads of a body, but Lemma 8.2 bounds them by $O(D)$ events while that body is active. The other two kinds affect only one own point head, and there are $O(|H|)$ such events.

A repaired head is checked against *all current* pair closure conditions. Another invalidation therefore needs a later saturation event or selection of that edge. This remains true if another endpoint's repair deletes the queued edge: an already valid head can be deleted only after a new pair-spanning event, or after selection. Static predicates are checked at both endpoint head generations and cannot later invalidate an otherwise valid head. Thus there are $O(Dp + |H|)$ head repairs and $O(\ell p)$ rebuilding point visits. Binary heaps give the logarithmic term, since $|H| \le \ell qD + f$. More explicitly, charge a stale point head to the first event since its last valid installation that invalidated that record. A rank or neighbor-pin saturation is charged at most once per point before that point's next repair, and there are $O(D)$ such events per active body. An own-pin or physical-pair saturation affects just its own point in each endpoint body. Selection of an original edge is charged to at most its two queued copies. Rebuilding a body's heads on growth of B_D accounts separately for at most ℓ visits to each of its points. Deleting several records during one repair does not create an event or a heap operation for each record.

Locally skipped original incidences are bounded by Lemma 8.1. A long safe-deletion chain charges each edge only its constant-time unlink; it has only one final heap update. Known-list cursor work sums to $O(f)$.

The heaps and known stream always expose the earliest edge eligible at an endpoint and open in its pair. An original edge skipped at both endpoints is already rejected by all layers there. Monotone flats cannot make it eligible later. An earlier eligible edge therefore cannot be hidden behind a later processed edge. Induction over selected edges gives exactly the reference sample and exactly the reference local layer contents. □

9 The fixed first-binding interface

Sampling and the sparse optimizer give the chronological basis F of $P_0 + H$. From it choose all current large blocks and the fixed exceptional old bodies X. Retire an exceptional body's identifier when that body is wholly absorbed by a large block. The active tracked tight sets intersect pairwise in at most one pin. Their total number of persistent identifiers is

$$R \le Q + |X| = O(q/D).$$

Lemma 6.1 applies to this family, including the exceptional bodies as identifiers present initially. It gives $O(R)$ known relationships and a set T of historically shared physical pins with $|T| \le 3\ell R$.

Lemma 9.1. *In $q^{1+o(1)}$ preprocessing time and $O(n)$ additional original-vertex work, the source history can be turned into the next stage with:*

1. *$q' \le C_* q/D$ disjoint mandatory tight frames;*
2. *$O(q')$ known chronological auxiliary edges;*
3. *one fixed body identifier $\lambda'(v)$ and one activation threshold $\tau'(v)$ per original vertex; an always inactive vertex uses $\lambda'(v) = \perp$.*

Here C_* depends only on k, ℓ. Closure in the new model is safe in the corresponding original weight prefix. An unselected original edge that fails its endpoint activation test is already spanned by earlier root-certificate edges.

Proof. Disjoint initial frames. Make one fresh tight frame for every tracked identifier. For every $z \in T$, also make a distinct fresh tight frame with exactly one used physical pin, namely z, and its k private attachments. The pin retains k free coordinates, because U_z has row rank k; the extra $\ell - k$ frame parameters impose no original constraint. Initial frames are disjoint and their internal bases are independent. There are

$$q' = R + |T| \le (1 + 3\ell)R \tag{15}$$

of them.

Known constraints. An actual attachment of a shared pin z to tracked frame A is represented by k edges from z to distinct core vertices of A. By (4), these impose $\xi_z = U_z \alpha_A$. To identify a retired frame B with frame A, use ℓ diagonal edges between their cores, with at most c copies per corresponding core pair. The width (5) is sufficient. Their degrees are at most $c < k$, their pair multiplicities at most c, and their number is ℓ. By (7), they are independent and make the two-frame union tight in the contraction by the two mandatory frames. Their constraints annihilate all motions with equal parameters. There are ℓ independent constraints on the 2ℓ parameters, so their nullspace is exactly that ℓ-dimensional equal-parameter space. They impose exactly $\alpha_A = \alpha_B$. No fresh merger pins are needed. Insert these known edges at the source F-prefix when the relationship becomes true. The compressed history has $O(R) = O(q')$ such edges.

Fixed bindings and times. For an original pin $v \notin T$, bind it to the *first* tracked identifier that contains it. If no such identifier exists, leave it unbound and inactive. This identifier is fixed for the whole next stage. Later retirement is handled by parameter equality, so no endpoint renaming is necessary. If v could belong to two distinct active tracked sets, the complete offline history would place it in T; then its separate free frame and known attachments retain the same physical coordinates. For $v \in T$, use that separate free frame as its fixed initial body.

Let $b(v)$ be the first tracked membership time; put $b(v) = +\infty$ when there is no membership. In the implementation $+\infty$ is the sentinel $m+1$, which is larger than every original query index. Compose the new threshold with the old one by

$$\tau'(v) = \max\{\tau(v), b(v)\}. \tag{16}$$

Frames that will be born later may already exist initially: no earlier usable original edge involves them, and their known relationships arrive only at the justified prefix. Coordinates of an original vertex bound to such a frame need not match the root coordinates before its activation time.

The chronological extension invariant. For each root weight prefix, every motion satisfying the selected original certificate has a lift to a motion of the current source model, with identical coordinates at all original vertices whose thresholds have passed. Each source model has its own coordinate space. It uses the same U_v and original-edge coefficients for an original physical identity v, but agreement of its coordinate vector with the root vector is imposed only after activation. Before activation the source coordinate may be chosen independently of the root coordinate. Thus "lift" does not assert simultaneous satisfaction of all stages' mandatory rows in one root coordinate space. The source rows are its mandatory initial bases, the known constraints arrived so far, and its own selected original sample rows so far. Original rows from ancestor stages are not copied into the new source model; the root motion already satisfies them.

To establish the invariant at the interface, first take a lift of the root motion to the parent model. Its F-prefix is a basis of its H-prefix. On each tracked tight set, the motion thus has form (4); choose that α_A for its new frame. For a persistent identifier its birth set already determines the parameter uniquely. Its growing sets and containing successor sets already present at this prefix have the same parameter, because they contain that birth set. Thus these choices are consistent with the arrived history. An identifier not yet born has no arrived relationship and its parameter can be chosen freely. Preserve the physical coordinates of shared pins. Attachments and parameter identifications hold at exactly their justified parent times. Unused future frame coordinates can be completed freely. If a root vertex is active in the child, it was active in the parent and already belongs to its tracked set, so its coordinates agree. Every sample edge selected subsequently at the child stage has active endpoints on arrival. Thresholds are fixed within a stage, so those rows continue to be evaluated on root coordinates in all later prefixes and are satisfied by the root certificate motion. This proves the invariant and hence the safety of source closure.

Activation discards. Consider an original edge outside H that passed its old thresholds but has an endpoint not yet in any tracked set. If it were open in the parent P_0+F-prefix, Lemma 6.3 would place both endpoints in the tracked family. It is therefore already parent-prefix-spanned. The old invariant still identifies its coordinates with the root coordinates, so this discard is safe. Failure of an earlier threshold was already justified at an earlier interface.

Computing vertex metadata. Only $O(q)$ physical pins occur in actual F-edges. A never-used lazy pin cannot partially enter a new maximal block: without new incident edges, its unique maximal block contains its whole old body. Such pins therefore share their old body's first-membership event. Registered-pin exceptions and T are obtained from the $O(q)$-size actual source history. For each original vertex, its old identifier and optional registered-pin record determine its new identifier and threshold. One $O(n)$ vertex pass computes all metadata without reading original edges. The sparse history preparation costs $q^{1+o(1)}$. □

Remark 9.2. The child need not inherit the closure of every small parent block. Such an edge may be open in the weaker child model and may be selected into a later sample. This is allowed: each stage selects at most $\ell qD+f$ original and known edges, and final optimization is performed in the original matroid. The argument uses safe discards, not equivalence of all parent and child closure answers.

10 One fixed algorithm and its total cost

10.1 Fixing the constants once

Use the constants

$$\begin{aligned} P_* &= 2\ell + \ell^2(1 + 2\ell/c), \\ A_* &= 16\ell^2/c, \\ C_R &= A_* + 2P_* + (2\ell/c + 3\ell)A_*, \\ C_* &= (1 + 3\ell)C_R, \end{aligned}$$

and fix

$$D = \lceil \max\{4\ell, 4C_*\} \rceil . \tag{17}$$

The threshold $s = \max\{2, \lfloor D/(4\ell) \rfloor\}$ satisfies $s \geq D/(8\ell)$. Thus $Q \leq A_* q/D$. The potential exceptions contribute at most $2P_* q/D$, and the historical straddlers at most $(2\ell/c + 3\ell)Q$. It follows that $R \leq C_R q/D$, and Lemma 9.1 gives

$$q' \leq C_* q/D \leq q/4. \tag{18}$$

The integer D depends only on k, ℓ, not on n, m, recursion depth, or an exponent-error parameter.

10.2 The initial stage

Give each original vertex v a disjoint constant-size tight frame with v as its one used physical pin, attached privately by k edges. Start with $q_0 = n$, $\tau_0(v) = 0$, and no known auxiliary edge. This introduces no restriction on original coordinates: one used pin retains k free coordinates in its ℓ-parameter body. Equivalently, in (6) every touched body contributes k, so the induced edge matroid is exactly the original (k, ℓ)-sparsity matroid. Initial auxiliary rows are not returned as original edges.

At stage j, the metadata are fixed. An original $e_i = uv$ is usable precisely when $i \geq \tau_j(u), \tau_j(v)$. All ancestor activation conditions have already been folded into these two thresholds. Only two endpoint lookups are performed; the edge does not query a sequence of ancestor closure indices. Within-body edges are spanned by the mandatory model. Static usability and within-body tests are applied during implicit head generation. Edges at vertices whose views never become active need not be discovered until the final array pass.

Known constraints carry root-prefix timestamps. Such a constraint may be justified by several selected original edges; its timestamp identifies the last possible supporting original position, rather than claiming that one original edge alone implies the constraint. Every positive such timestamp is the index of an original edge selected at an ancestor or earlier stage and already removed from the live lists. The same is true of every positive first- membership and activation timestamp. Constraints with that timestamp are ordered after that original edge and before the next original edge. An inherited known-constraint time has the same property transitively. Thus no still-live original query at the same root index precedes a constraint it uses. Auxiliary edge elements still have distinct identifiers. For a still-live original e_i, passing a finite positive activation threshold therefore means that threshold is strictly less than i. All source conditions used to discard e_i are valid in the root prefix strictly preceding e_i, not merely in a prefix that includes the queried edge. Timestamp zero refers to the unrestricted initial model. Multiple auxiliary edges at one timestamp have a fixed internal order; original queries at that timestamp have already been removed.

Algorithm 1 Uniform deterministic original-edge certificate

Input: Ordered original edges; fixed $k < \ell < 2k$.
Output: $\mathcal{C}$ with $|\mathcal{C}| = O(n)$ and $B_\ell(\mathcal{C}) = B_\ell(E)$.

1: Fix D by (17).

2: Keep only the first c copies of every physical pair.

3: Build the shared ordered incidence lists once.

4: $\mathcal{C} \leftarrow \varnothing$; initialize the n-body model.

5: **While** the current body count q exceeds D:

   a. Generate H by the implicit adaptive-pair sampler.

   b. Add every selected original edge to $\mathcal{C}$.

   c. Permanently unlink selected or safely spanned original edges.

   d. On the explicit sparse graph, compute the chronological basis F after the mandatory P_0.

   e. Compute logical block history, X, tracked identifiers, and shared pins T.

   f. Build the child model and fixed vertex metadata by Lemma 9.1.

6: In one final pass over the original edge array, process still-live usable edges in the constant-size source model, merging the known-constraint stream chronologically.

7: Add its rank-increasing original edges to $\mathcal{C}$.

8: **Return** $\mathcal{C}$.

10.3 The constant-body terminal stage

When $q \le D$, the source has constant total body parameter dimension. Use the incremental component matrix of the component pebble-game algorithm [11], on a preallocated constant-size vertex universe containing all possible successful registrations. Pre-register the endpoints of the $O(q)$ known rows, as well as the cores. Reserve at most $2\ell q$ additional original-pin slots. There are at most ℓq independent new source rows and $O(q)$ known rows, and consequently only $O(q)$ actually registered query endpoints.

If $q = 0$, all remaining original vertices are inactive and the final pass only performs the activation test. For $q > 0$, apply both-endpoint activation and within-body tests first. An unsuccessful virtual-pin query requires no registration:

1. For two registered endpoints, read their common-component bit.

2. For a registered u and an unregistered pin attached to frame B, test the closure bits from u to the k distinguished core vertices of B. They are all true exactly when u belongs to the maximal block containing B.

3. For two unregistered pins of frames A, B, test all cross-pair core closure bits. They are all true exactly when the two frames lie in one maximal block.

The frame size is constant. For the second equivalence, the k tested edge rows, together with a tight basis of B, form a tight graph on $B \cup \{u\}$ (membership $u \in B$ is immediate). If all tested rows are spanned, Lemma 2.2 puts u and B in one block. The converse is Lemma 2.1. For the third equivalence, all true cross-pair bits span the ℓ diagonal rows used to join A and B; those rows and the two frame bases form a tight graph. Apply Lemma 2.2 again. Finally a pin with no new incident edge belongs only to the maximal block containing its frame, by the fresh-pin

argument in Section 5. These statements also show that omitting unsuccessful registrations leaves all answers on registered vertices unchanged. On a successful original query, register its missing endpoints with private attachments and insert the original edge. The bounded total number of insertions costs $O(1)$ for fixed k, ℓ, D, and each remaining original query has constant cost. The final pass therefore costs $O(m+n)$.

10.4 Correctness and the uniform additive bound

Proof of Theorem 1.2. For $\ell \le k$, use Proposition 4.2. For the upper range, construct $\mathcal{C}$ by Algorithm 1 and then run $\textsc{Sparse}(\ell, \mathcal{C})$ in the original matroid.

Correctness. The initial physical-pair truncation preserves ordered greedy prefixes, since each later copy is spanned by the first c copies. On the retained input, every later original discard has a certificate using only earlier selected original edges in its own weight prefix. Pair-filter discards use earlier H_{ab}. Activation discards and within-body discards are justified by Lemma 9.1. The motion-extension invariant transfers every source dependence to the original matroid. Known auxiliary conditions themselves are supported by earlier original samples, transitively; they are not included in the original output.

A weaker child model may select an original edge that was already dependent in the parent. Keeping such an extra candidate is harmless: all samples are retained in $\mathcal{C}$, and final optimization uses the original matroid. At the terminal stage, the exact source basis and its known constraints justify all remaining discards in the same way. Thus every omitted original edge is spanned by the earlier $\mathcal{C}$-prefix. This preserves every prefix rank and hence the ordered greedy output. The final sparse optimization returns the required maximum-weight (k, ℓ)-sparse subgraph. Initial truncation witnesses also transfer to the final certificate by this prefix-preservation property.

Original input work. Reading and truncating the input and building the incidence lists costs $O(m+n)$. Let d_j count original unlinks at stage j. Each edge is unlinked at most once, so

$$\sum_j d_j \le m.$$

Its two list modifications take constant time. Lemma 8.3 does not attach a separate heap update to every deleted edge. The single final array pass adds $O(m)$. Original records skipped locally without deletion are charged to the stage's point-head bound, rather than to a repeated complete edge scan. The direct original-edge work is therefore $C_{k,\ell} m$, with a coefficient independent of recursion depth.

Point and source work. By (18),

$$\sum_j q_j = O(n), \qquad \text{number of stages} = O(\log(n+2)).$$

Known-edge counts satisfy $f_j = O(q_j)$, so $\sum_j f_j = O(n)$. The explicit graph for a stage sample has $O(Dq_j + f_j)$ vertices and edges, including its constant frames and registered pins. After computing F, unused physical pins can again be treated lazily. Since D is fixed, all sparse optimizations and prefix-history constructions together cost

$$\sum_j (Dq_j + f_j)^{1+o(1)} = n^{1+o(1)}.$$

A stage handles at most the n original points and $O(q_j)$ known-edge and frame points. Thus $p_j \le n + O(q_j)$ and $\sum_j p_j = O(n \log(n+2))$. Vertex-metadata passes, head rebuilds, and the non-input part of (14) sum to $O(n \log^2(n+2))$. All freshly created auxiliary points over all stages number $O(\sum_j q_j) = O(n)$.

Finally,

$$|\mathcal{C}| \le \sum_j (\ell D q_j + f_j) + O(1) = O(n).$$

Its final sparse optimization takes $n^{1+o(1)}$ time. Combining the bounds gives

$$O\left(m + n\log^2(n+2) + \sum_j q_j^{1+o(1)}\right) = O(m + n^{1+o(1)}).$$

All heaps, local oracles, initializable tables, flows, decompositions, and history computations are deterministic. The constants were fixed once in (17), establishing the claimed uniform quantifier order. □

Remark 10.1. The generic matrices and motions occur only in the correctness proof. No random coordinates or polynomial identity tests are used. Body-pair tables may use quadratic address space with constant-time initialization. The claimed bound is the uniform additive running time.

11 Exact checks and their scope

The original manuscript reports the following exact combinatorial checks from its development. Their scripts and logs were not supplied with the manuscript, so these original counts have not been independently reproduced in this review.

- For fixed body partitions, 160 implicit-local sampling tests, with 72 533 input edges in total, produced exactly the same local layers and samples as direct chronological execution.
- The adaptive selected-pair filter was checked on another 160 tests with 120 414 input edges. All layer contents and selected edges agreed. These runs had 43 497 safe deletions and only 3516 lazy heap-head repairs. The saturation-event bounds were checked for every active body. A separate heavy-pair example had 3765 input edges, 33 selected edges, and 3732 safe deletions.
- Endpoint coverage was checked on four exact source graphs with 2014 input edges. A further structured 601-edge example had a genuinely open omitted edge: the sample contained the first 600 edges, and the last edge's endpoints were covered by two different large blocks.
- Fixed first bindings, separate free shared-pin frames, and timed diagonal identifications were checked on 24 exact histories. There were 4046 helper queries without an external parent-closure guard, including 952 edges already spanned in the parent. Another 172 edges were safely omitted by activation times before their future frames were born. Every resulting original-edge certificate gave the same ordered greedy optimum as the full input.

The October 11, 2026 review additionally supplies reproducible Python scripts using integer arithmetic, exhaustive vertex-subset tests, and integral minimum cuts. With fixed seeds 20261011 and 20261012, all of the following independent checks passed:

- 2158 weak/strong circuit and greedy-compatibility instances, including both edge orders on every four-vertex multigraph with zero, one, or two copies per pair for $(k, j) = (2, 3)$; 1429 prefix histories and 83 992 closure queries.
- 1800 disjoint-body independence queries against exhaustive physical-vertex sparsity; 250 adaptive-sampling instances with 43 093 input edges and 1040 separate pair-discard safety checks.
- 250 implicit/reference sampling comparisons, with 43 686 original edges and 2189 timed known rows. Samples and every local layer agreed. The runs checked 26 460 safe unlinks and 7665 stale-head repairs. The test uses a flat heap of point heads, which exposes the same minimum as the nested heaps; it does not measure their runtime.

- 200 reverse rooted-cut histories with 1485 states, compared with exhaustive cut enumeration.
- 250 lower-range union-certificate instances, including 188 cases with $\ell = 0$ or $\ell = k$. The final-present-type shortcut chose the same layer as direct execution, and every certificate preserved the full ordered greedy output.
- 100 covered tracked-frame interface histories, involving 278 identifiers, 132 historically shared pins, 1096 known-row queries and 555 remaining original queries. All 441 discards, including 85 activation discards, were safe in the strict parent prefix, and every certificate preserved the ordered parent greedy output. These tests augment the exceptional family to enforce qualitative endpoint coverage; they do not test its asymptotic size bound.
- 3796 terminal virtual-pin closure queries, compared with the exact body-count oracle after explicit virtual-pin inclusion.

The scripts and their machine-readable result files accompany the reviewed source. These finite checks exercise specific lemmas and interfaces. They are not a full implementation, formal verification, or an empirical verification of the asymptotic running time. The running-time claim rests on the amortized proof of Theorem 1.2 and the cited algorithmic primitives.

References

[1] Torben Hagerup and Frank Kammer. *On-the-Fly Array Initialization in Less Space.* ISAAC 2017, LIPIcs 92, 44:1–44:12. `https://doi.org/10.4230/LIPIcs.ISAAC.2017.44`.

[2] Ileana Streinu and Louis Theran. *Natural Realizations of Sparsity Matroids.* Ars Mathematica Contemporanea 4(1), 141–151, 2011. See Section 3 for the generic U construction. `https://arxiv.org/abs/0711.3013`.

[3] Jan van den Brand, Li Chen, Rasmus Kyng, Yang P. Liu, Richard Peng, Maximilian Probst Gutenberg, Sushant Sachdeva, and Aaron Sidford. *A Deterministic Almost-Linear Time Algorithm for Minimum-Cost Flow.* FOCS 2023; arXiv:2309.16629. `https://arxiv.org/abs/2309.16629`.

[4] Pavel Arkhipov and Vladimir Kolmogorov. *Faster algorithms for packing forests in graphs and related problems.* arXiv:2409.20314, version 3, 2026. `https://arxiv.org/abs/2409.20314`.

[5] Bence Deák and Péter Madarasi. *Asymptotically Faster Algorithms for Recognizing (k, ℓ)-Sparse Graphs.* arXiv:2604.13025, version 2, 2026. `https://arxiv.org/abs/2604.13025`.

[6] Anand Bhalgat, Ramesh Hariharan, Telikepalli Kavitha, and Debmalya Panigrahi. *Fast Edge Splitting and Edmonds' Arborescence Construction for Unweighted Graphs.* SODA 2008, pp. 455–464. `https://users.cs.duke.edu/~debmalya/papers/soda08-splitting.pdf`.

[7] Loukas Georgiadis, Dionysios Kefallinos, Anna Mpanti, and Stavros D. Nikolopoulos. *An Experimental Study of Algorithms for Packing Arborescences.* SEA 2022, LIPIcs 233, 14:1–14:16. `https://doi.org/10.4230/LIPIcs.SEA.2022.14`.

[8] Harold N. Gabow. *The minset-poset approach to representations of graph connectivity.* ACM Transactions on Algorithms 12(2), 24:1–24:73, 2016. `https://doi.org/10.1145/2764909`.

[9] Loukas Georgiadis, Evangelos Kipouridis, Evangelos Kosinas, Charis Papadopoulos, and Nikos Parotsidis. *Computing the* $(k+2)$*-Edge-Connected Components in* k*-Edge-Connected Digraphs in Subquadratic Time*. ICALP 2026, LIPIcs 374, 95:1–95:23. See Proposition 27 for the minset-poset construction bound. `https://arxiv.org/abs/2604.27474`.

[10] Harold N. Gabow and Robert Endre Tarjan. *A linear-time algorithm for a special case of disjoint set union*. Journal of Computer and System Sciences 30(2), 209–221, 1985. `https://doi.org/10.1016/0022-0000(85)90014-5`.

[11] Bence Deák and Péter Madarasi. *Quadratic-Time Algorithm for the Maximum-Weight* (k, ℓ)*-Sparse Subgraph Problem*. arXiv:2511.20882, 2025. `https://arxiv.org/abs/2511.20882`.